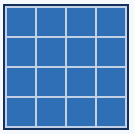
Three forms, one factorization

All three reconstruct A exactly. They differ in how much of U and Σ is kept — and what is dropped is precisely the part that multiplies against zero singular values.

3
FORMS

EVERYTHING KEPT 1



Full SVD | U and V both square and orthogonal

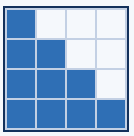
SHAPES	WHAT IS KEPT	WHAT IS DROPPED
$U : m \times m$, $\Sigma : m \times n$, $V : n \times n$	all of U, V, and a rectangular Σ	nothing

USED FOR

theory, the four subspaces

The only form where U and V are genuinely orthogonal matrices, so $UU^{\mathsf{T}} = I$ as well as $U^{\mathsf{T}}U = I$. That is what makes it the form to state results with — the last $m - r$ columns of U span the left null space and would otherwise be gone.

TRIMMED TO WHAT MULTIPLIES 2



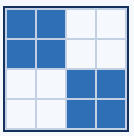
Thin SVD | $m > n$ — keep n columns of U

SHAPES	WHAT IS KEPT	WHAT IS DROPPED
$U_n : m \times n$, $\Sigma_n : n \times n$, $V : n \times n$	the first n columns of U	the last $m - n$ columns of U

USED FOR

least squares on tall matrices

The dropped columns multiply rows of Σ that are entirely zero, so removing them changes nothing about the product. Note U_n is no longer square: $U_n^{\mathsf{T}}U_n = I_n$ still holds, but $U_nU_n^{\mathsf{T}}$ does not.



Compact SVD | keep only the r nonzero singular values

SHAPES	WHAT IS KEPT	WHAT IS DROPPED
$U_r : m \times r$, $\Sigma_r : r \times r$, $V_r : n \times r$	the rank-r content only	every zero singular value and its vectors

USED FOR

storage, rank-deficient matrices

The most economical exact form: Σ_r is square with a strictly positive diagonal, so it is invertible — which is what makes the pseudoinverse computable as $V_r\Sigma_r^{-1}U_r^{\mathsf{T}}$.

Nothing is approximated anywhere in this table. The full form carries orthonormal bases for all four fundamental subspaces, which is why it is the one to state theorems with; the compact form carries only what contributes to A, which is why it is the one to store. Approximation begins only when the compact form is truncated below rank r.